MACHINE LEARNING PROJECT · 2026

Food Delivery Time Prediction

Predicting how many minutes a food order will take to arrive, using distance, weather, traffic and more.

Group no. 34

Team & roles

Member	Roll No.	Role
Divij Anghan	2026AIB1012	**Presentation lead**
Mahak	2026MCB1540	**Documentation lead**
Nanong Tamut	2026DAB1298	**Video lead**
Aavik	2026EEB1313	**Execution lead**

GitHub repository

github.com/aavikrao/Food-Delivery-Time-Prediction

Commit history on the main branch

2 Problem and data

Why delivery time is hard to guess, and what we trained on

What and why

Food Delivery Time Prediction using Machine Learning.

Customers want a reliable ETA and platforms need to plan couriers. Distance alone is not enough: weather, traffic, time of day and kitchen prep time all push the delivery time up or down.

Dataset source

Food_Delivery_Times.csv

Public GitHub dataset, repository sanskar770 / Delivery_Time_Prediction.

Open the CSV on GitHub

Size and columns

1,000 rows

9 columns

The 9 columns

Order_ID — identifier	Distance_km — numeric	Weather — categorical
Traffic_Level — categorical	Time_of_Day — categorical	Vehicle_Type — categorical
Preparation_Time_min — numeric	Courier_Experience_yrs — numeric	Delivery_Time_min — TARGET

3 Approach

Pipeline: data → model → output

DATA
CSV dataset
- 1,000 rows × 9 columns
- Order_ID dropped (no signal)

PREPARE
Clean and encode
- Check missing values
- Encode Weather, Traffic, Time, Vehicle
- Scale numeric columns

SPLIT
Train / test
- Hold out a test set
- Model never sees it while training

MODEL
Regression
- Fit on training set
- Target: Delivery_Time_min

OUTPUT
Prediction
- Estimated delivery time in minutes
- Scored on the test set

Libraries used

pandas
load and clean data

NumPy
numeric operations

scikit-learn
preprocessing, model, metrics

Matplotlib
charts

4 Results

Linear Regression beat Random Forest on every metric



Most orders take roughly 30–80 min; a few exceed 100 min



Longer distances mean longer deliveries, with more spread

MAE

6.06

minutes

RMSE

8.95

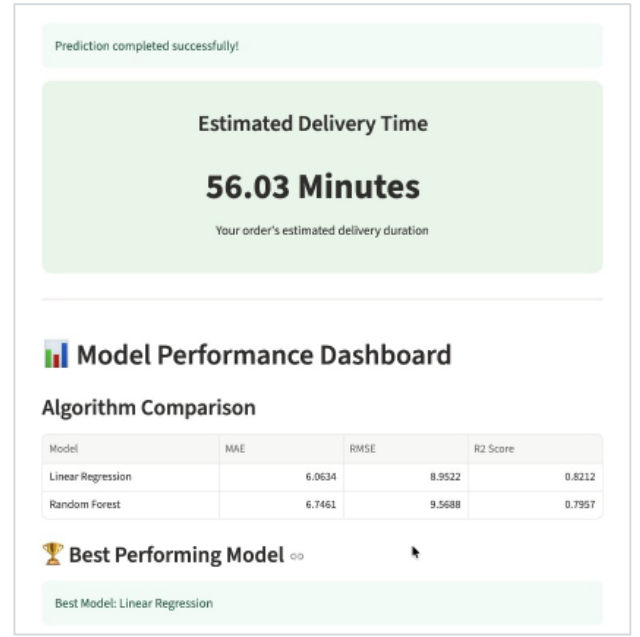
minutes

R^2 score

0.82

test set

Demo screenshot



Live Streamlit demo: prediction and model dashboard

Predicted 56.03 min · Best model: Linear Regression

5 Learnings

What we took away, who did what, and how AI was used

What worked, what didn’t

Worked

- Compact, tidy dataset: 1,000 rows, 8 usable features
- Mix of numeric and categorical inputs gave a realistic regression problem

Didn’t / limits

- Only 1,000 rows, so results can swing between splits
- Real deliveries have more factors than these columns capture

Who did what

Divij Anghan
Presentation lead
Slides and presenting

Mahak
Documentation lead
Report and write-up

Aavik
Execution lead
Code, model and repo

Nanong Tamut
Video lead
Project video

AI-use disclosure

AI tools were used to assist with project development and documentation.

ChatGPT

Used for:

- Project-related guidance
- Understanding Machine Learning concepts and algorithms
- Troubleshooting coding errors
- Assistance with project documentation and README preparation

Verification

Python scripts were executed by the team, and generated model evaluation results were checked against the output CSV file.

The team is responsible for understanding, reviewing, testing, and explaining the complete project.